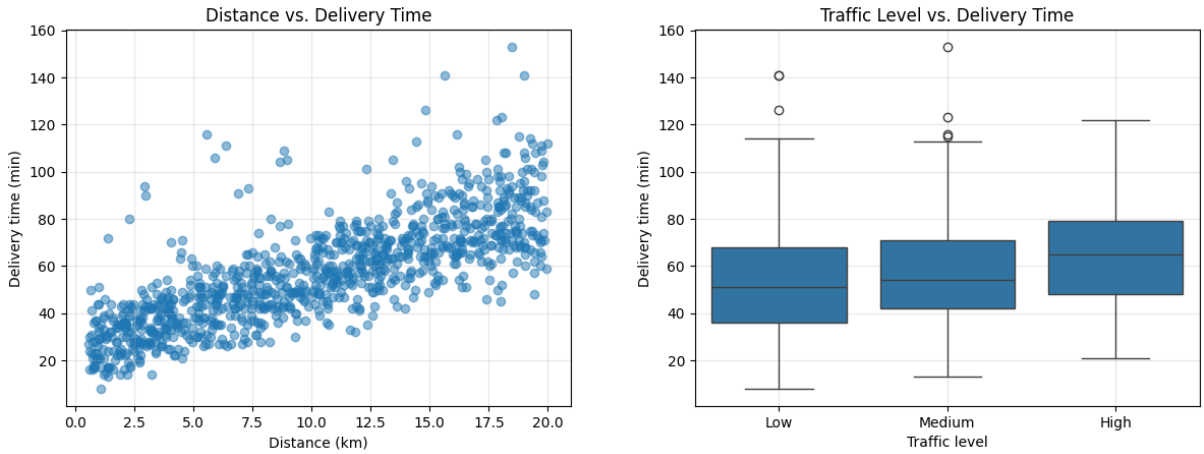
# CS461 – Assignment 2

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For a food delivery app, informing users with accurate delivery time estimates could be the difference between a satisfied or disgruntled customer. Delivery time can influence customer repurchase behavior in quick commerce[1], and late deliveries weigh more heavily on repeat purchases than equally early ones. Furthermore, negative experiences may have a stronger effect on customer behavior than equally sized positive experiences[2]. However, estimating delivery times is no easy task, as it may be affected by a multitude of factors, including traffic, weather, distance, and courier characteristics [3]. In this assignment, an Ordinary Least Squares (OLS) Linear Regression model will be trained and evaluated using food delivery data to predict delivery times.

Linear regression is a good fit for this problem, because our target, delivery time, is a continuous feature and our suspected main predictors, distance and traffic have a simple linear relationship with the target where each increase in one increases delivery time linearly, as we can see in Figure 1. Figure 1 (a) shows the linear relationship between Distance and Delivery Time, while Figure 1 (b) shows a linear relationship between traffic level and delivery time.




(a) Delivery time vs. distance

(b) Delivery time vs. traffic level

Figure 1: Relationship between delivery time and its two main predictors

## The data

The dataset I will explore in this assignment is the Food Delivery Time Prediction dataset from Kaggle. The dataset has a total of 9 features. our target is delivery time in minutes and the

features I suspect would have the most impact are distance, traffic level and weather in that order.

| Name | Meaning | Possible Values |
|---|---|---|
| Order_ID | Unique ID for each order | Numeric |
| Distance_km | The delivery distance in kilometers | Numeric |
| Weather | The weather conditions during delivery | Clear, Rainy, Snowy, Foggy, Windy |
| Traffic_Level | The level of traffic during delivery | Low, Medium, High |
| Time_of_Day | The time when the delivery took place | Morning, Afternoon, Evening, Night |
| Vehicle_Type | Type of vehicle used for delivery | Bike, Scooter, Car |
| Preparation_Time_min | The time required to prepare the order | Numeric |
| Courier_Experience | Experience of the courier in years | Numeric |
| Delivery_Time_min | The total delivery time in minutes | Numeric |

Table 1: Dataset features and their descriptions

In this assignment, the dataset will be used to train an OLS Linear Regression model to predict delivery times. Its performance will be evaluated, and then Ridge and Lasso regression will be used to investigate whether regularization can improve the results.

The dataset contains a total of 1000 rows, 117 contain at least one missing value. Removing these rows would leave 883 rows only, so I decided to impute the missing values instead. The missing values in `Courier_Experience_yrs` were replaced with the median because they are continuous values, while those in `Weather`, `Traffic_Level` and `Time_of_Day` were replaced with their modes because they are categorical. 6 outliers were found in delivery_time but I see no evidence that they are invalid.

Order_ID was removed because it is a unique identifier. Traffic_Level was mapped to numerical values, with Low represented by 0, Medium by 1 and High by 2, the remaining categorical features were one-hot encoded because their categories do not have a natural numerical order.

A MinMaxScaler was used to scale the features to a range between 0 and 1. This is important for regularized models such as Ridge and Lasso because they penalize coefficient magnitudes, and features with different scales could otherwise be penalized unfairly. The scaler was fitted on the training data and then applied to the test data to avoid data leakage.

## Baseline model and evaluation

An OLS Linear Regression model was fitted on the training data and evaluated on both the training and testing sets.

As seen in figure 2, the model performs well and is not far off from the perfect prediction line.

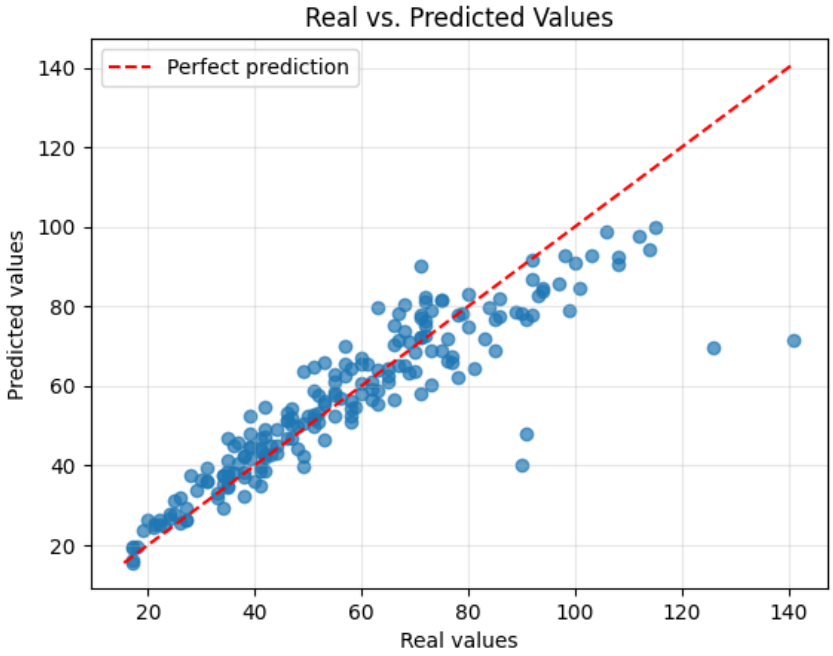



Figure 2: Enter Caption

As seen in table 2, the test MAE of 6.73 minutes means that on average,the predicted delivery time is off by about 6.7 minutes from the real one, the test RMSE is higher at 10.69 minutes, because RMSE penalizes large errors more than small ones, so the difference between the two shows that some predictions have much bigger errors than the average. For comparison the standard deviation of the delivery time in the data is about 22 minutes, so the errors of the model are clearly smaller than the natural spread of the target.

To check for overfitting the training and testing errors were compared. The testing MAE is only 0.26 minutes higher than the training MAE and, the testing RMSE is also only 0.26 minutes higher than the training RMSE. Such a small gap means the model performs almost the same on data it has not seen, so the baseline model is not overfitted.

| Metric | Training | Testing |
|---|---|---|
| MAE (min) | 6.47 | 6.73 |
| RMSE (min) | 10.43 | 10.69 |

Table 2: Baseline OLS performance

## Model improvements and hyperparameter tuning

Ridge (L2 penalty) and Lasso (L1 penalty) regression were implemented to see if regularization could improve the baseline results. With the default alpha = 1.0 Ridge performed almost the same as OLS, while Lasso performed clearly worse. Therefore alpha was tuned using GridSearchCV with 5 fold cross validation over 30 values between 0.001 and 100 spaced on a log scale and the scoring metric was MAE.

GridSearchCV works by trying every alpha value in the grid, for each value the training data is split into 5 folds, the model is trained on 4 folds and validated on the remaining one and this is repeated so every fold is used for validation once. The alpha with the best average validation score is selected. Its advantages are that it is simple, it checks every value I give it and it uses cross-validation so the choice does not depend on one lucky split. Its disadvantages are that the

cost grows quickly with the number of values and folds and it can only choose from the values in the grid, so the true best value may fall between them or outside them.

| Model | Best alpha | Train MAE | Test MAE | Train RMSE | Test RMSE | Search time (s) |
|---|---|---|---|---|---|---|
| OLS | – | 6.4715 | 6.7336 | 10.4331 | 10.6935 | – |
| Ridge (tuned) | 0.001 | 6.4716 | 6.7336 | 10.4331 | 10.6936 | 0.73 |
| Lasso (tuned) | 0.001 | 6.4720 | 6.7349 | 10.4331 | 10.6951 | 0.29 |

Table 3: Comparison of OLS and tuned Ridge and Lasso

Tuning had no measurable effect on the models, both Ridge and Lasso chose an alpha of 0.001 which is the smallest value. The coefficients also remained unchanged as seen in figure 3. The search took under one second but since it had no measurable effect on the models, I would say tuning is not worth it and the baseline OLS model is enough.

In figure 2 we can see the coefficient value for each feature, we can see that OLS (blue) Ridge (orange) and Lasso (green) have the same coefficients, which means the tuning did not alter the model, we can also see that distance is the feature that matter the most in predicting our target of delivery time, since it has the biggest coefficient, in contrast, the coefficients for Time_of_Day_Night and Time_of_Day_Morning is almost zero, meaning it has almost no affect on the target, while Courier_Expeirence_yrs has a negative coefficients, which suggests that less experienced couriers make deliveries faster than more experienced ones, which is an interesting observation that begs a few of question I will not attempt to answer.

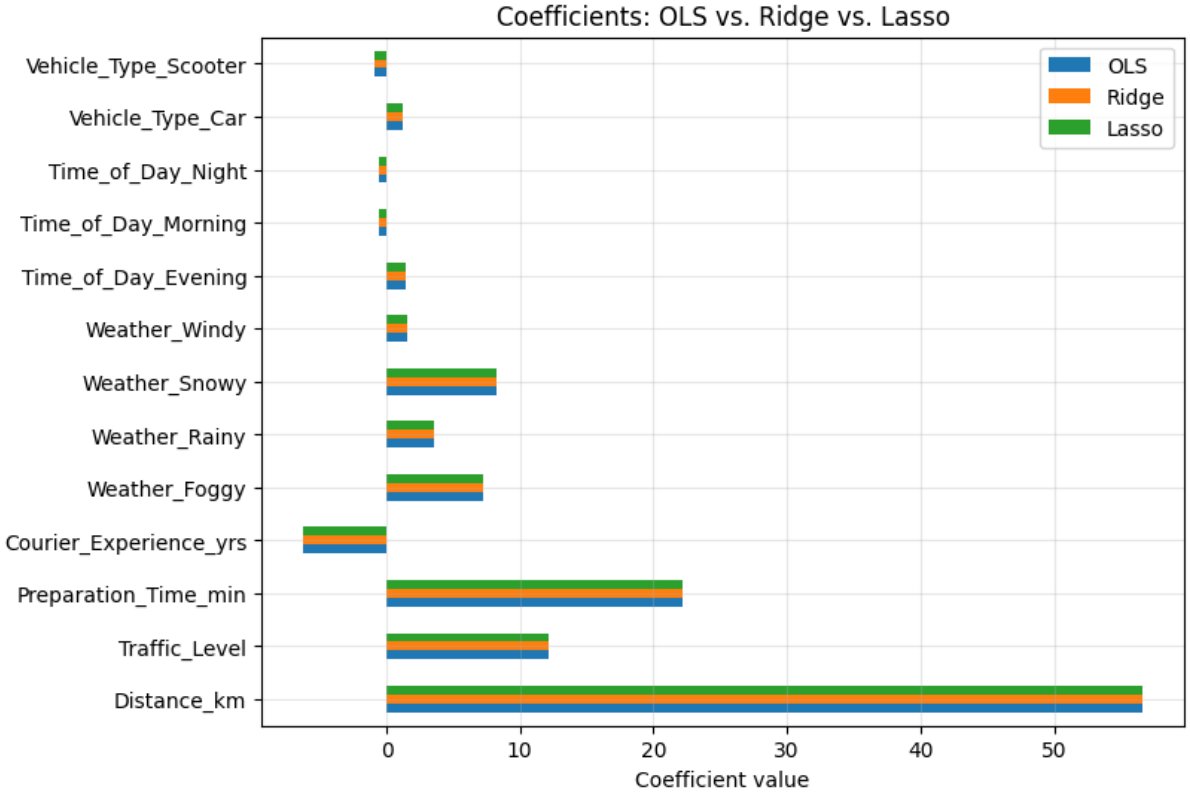



Figure 3: Comparison of Coefficients

## References


| No. | Source |
|---|---|
| [1] | Harter et al. (2024), Journal of Service Research (SAGE) |
| [2] | Baumeister et al. (2001), Review of General Psychology, 5(4), 323–370 |
| [3] | Chang (2025); Kalipaksi et al. (2026), Proceedings of ICEMCI 2025 (Atlantis Press) |


Table 4: References cited in this report

The notebook used for this assignment is available at https://Github.com/CollCaz/Assignment_Notebook